

AI Sales Training Platform



Complete Technical Manual & API Documentation

Platform Version: 1.1.0

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Core Architecture Pillars Included:

- Multi-Agent AI Orchestration & Live Conversation Flow
- Speech-to-Text (Whisper) & Real-time Text-to-Speech (ElevenLabs WS)
- Automated Sales Methodology Coverage (MEDDIC, BANT, SPIN, etc.)
- Manual Opportunities CRUD API Integration
- Scraping Engines, Data Scrapers & Performance Scoring Metrics

1. Technology Stack Overview

The AI Sales Training Platform is built using a modern, asynchronous Python architecture. The system is designed for ultra-low latency audio processing, high-concurrency real-time communication via WebSockets, and deep analytical capabilities using generative AI models.

Technology	Module / Library	Role in System
FastAPI	fastapi==0.109.0	Core Async Web Framework
Uvicorn	uvicorn==0.27.0	ASGI Production Web Server
MongoDB	pymongo & motor	Async Database Persistence
OpenAI GPT	openai>=1.12.0	GPT-4o Multi-Agent Orchestration
ElevenLabs	elevenlabs>=1.2.2	WebSocket / HTTP Voice Synthesis
Whisper	OpenAI API / Whisper	Speech-to-Text Audio Transcription
AWS S3	boto3==1.34.34	Secure Cloud Audio Segment Archival
FFmpeg	imageio-ffmpeg==0.6.0	On-the-fly Audio Recoding & Merger

Core Architectural Layers

- **Client Layer:** The front-end browser interacts with the backend APIs via standard HTTP REST requests and establishes a persistent stateful WebSocket connection for live-voice simulations.
- **REST API Router Layer:** FastAPI router handlers organize endpoint paths by domain (salesperson, company, meeting, conversation, admin, opportunities).
- **Service Layer:** Orchestrates API calls to external services. Employs background processing to execute non-blocking operations like audio upload, transcription, and post-session analytics.
- **Database & Storage:** MongoDB collections store profiles, meeting settings, transcripts, and analytics. AWS S3 stores audio recordings (.mp3/.webm) for each turn to maintain a record of sessions.

2. System Configuration & Deployment

For the platform to function properly, several API keys and cloud storage buckets must be configured. This configuration is managed via environment variables stored in a local '.env' file.

Required Environment Variables

```
# External AI Service Keys
OPENAI_API_KEY=sk-proj-... # GPT-4o and Whisper access
ELEVENLABS_API_KEY=sk_... # High-quality voice synthesis
GOOGLE_API_KEY=AIZA... # Google search API for company research

# MongoDB Database Config
MONGODB_URL=mongodb+srv://user:pass@cluster.mongodb.net
MONGODB_DB_NAME=julienmts-voice-sales

# Cloud Storage (AWS S3)
AWS_ACCESS_KEY_ID=AKIA...
AWS_SECRET_ACCESS_KEY=...
AWS_REGION=eu-north-1
S3_BUCKET_NAME=salesman-practice-audio
```

How to Run the Application

To start the application locally in a development environment, follow these commands:

```
# 1. Install all system dependencies
pip install -r requirements.txt

# 2. Configure Environment
cp .env.example .env
# Open and edit .env to insert proper credentials

# 3. Start Backend server
uvicorn main:app --reload --host 0.0.0.0 --port 8000

# 4. Access Interactive Documentation
# Open your browser and navigate to:
# - Swagger UI: http://localhost:8000/docs
# - ReDoc:      http://localhost:8000/redoc
```

To run the front-end simulation test tool, simply open the HTML client in any browser:

```
# Locate the test client file in the root directory:
```

```
# path: ./voice-conversation-client.html  
# Double-click or open it in Google Chrome/Firefox to start testing.  
# Simply enter a meeting ID and click "Connect & Start".
```

3. API Reference & Analysis

This section outlines key endpoints in the application, including the new manual opportunities CRUD API. All REST endpoints utilize standardized JSON bodies and return consistent responses.

Manual Opportunities API (CRUD)

This new set of endpoints allows users to manually input and manage business opportunities independent of AI-generated insights. The opportunities are stored in a dedicated 'opportunities' MongoDB collection.

- **POST /api/opportunities/** - Creates a new manual opportunity record linked to a company.
- **GET /api/opportunities/company/{company_id}** - Retrieves all manual opportunities for the specified company.
- **GET /api/opportunities/{opportunity_id}** - Fetches a single opportunity details.
- **PUT /api/opportunities/{opportunity_id}** - Updates an opportunity (supports partial updates).
- **DELETE /api/opportunities/{opportunity_id}** - Permanently removes an opportunity.

POST /api/opportunities/ - Sample Request Body

```
{
  "company_id": "60d5ec49f83f2c31e8a4a7b1",
  "name": "Enterprise Software Licensing Tier 3",
  "value": "$150,000",
  "stage": "Proposal",
  "close_date": "Dec 15, 2026",
  "probability": 65
}
```

GET /api/opportunities/company/{company_id} - Sample Response Output

```
{
  "success": true,
  "data": [
    {
      "_id": "665e8a719c8f1025a1b3c4d5",
      "company_id": "60d5ec49f83f2c31e8a4a7b1",
      "name": "Enterprise Software Licensing Tier 3",
      "value": "$150,000",
      "stage": "Proposal",
      "close_date": "Dec 15, 2026",
      "probability": 65,
      "created_at": "2026-06-04T12:00:00.000000"
    }
  ]
}
```

```
],  
  "message": "Opportunities retrieved successfully"  
}
```

4. Sales Methodology Coverage API

To analyze how effectively a salesperson follows framework guidelines, the platform provides a Methodology Coverage API. When a meeting completes, the system parses the entire session transcript and evaluates it against specific criteria (such as MEDDIC, BANT, or SPIN).

POST /api/conversation/{meeting_id}/methodology-analysis

This endpoint runs an LLM evaluation over the conversation history to map question-and-answer pairs to the framework parameters, calculates an overall coverage percentage score, and lists concrete improvements.

Sample API Output Response

```
{
  "success": true,
  "data": {
    "meeting_id": "meet_abc123xyz",
    "session_id": "session_8877",
    "methodology": "MEDDIC",
    "overall_coverage_score": 67,
    "fields_analyzed": [
      {
        "field": "metrics",
        "covered": true,
        "justification": "Salesperson asked about target growth rates and Bob responded that they want to achieve 25% YoY.",
        "question_asked": "What are your core growth metrics for next fiscal year?",
        "answer_given": "We're aiming for 25% YoY growth on our software product line.",
        "suggestions": "Great question. Follow up on what the current actual growth is to highlight the gap."
      },
      {
        "field": "economic_buyer",
        "covered": false,
        "justification": "The salesperson failed to identify the budget holder or verify Bob has final sign-off.",
        "question_asked": null,
        "answer_given": null,
        "suggestions": "Ask Bob directly who holds budget authority and what executive sponsors are involved."
      }
    ],
    "generated_at": "2026-06-04T12:05:00+06:00"
  },
  "message": "Methodology coverage analysis complete -- 67% coverage"
```

```
}
```

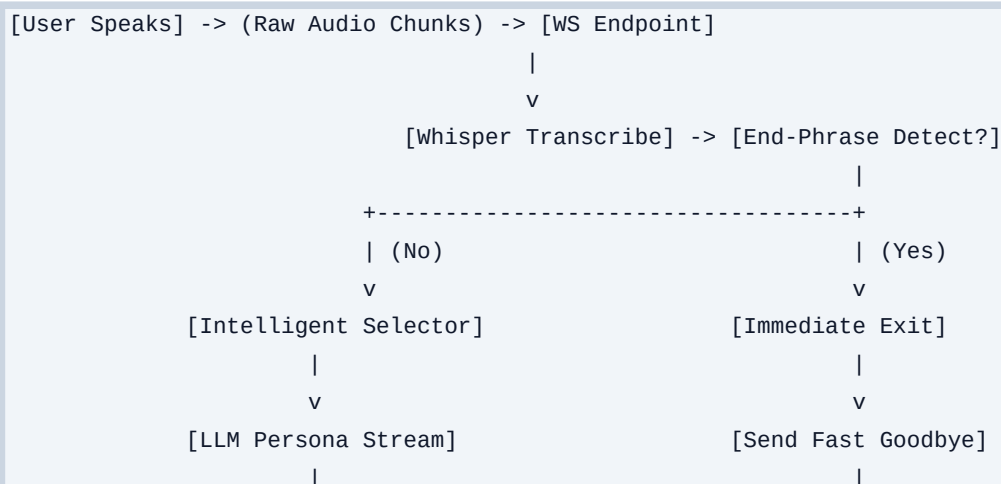

5. Real-Time Voice Simulation Workflow

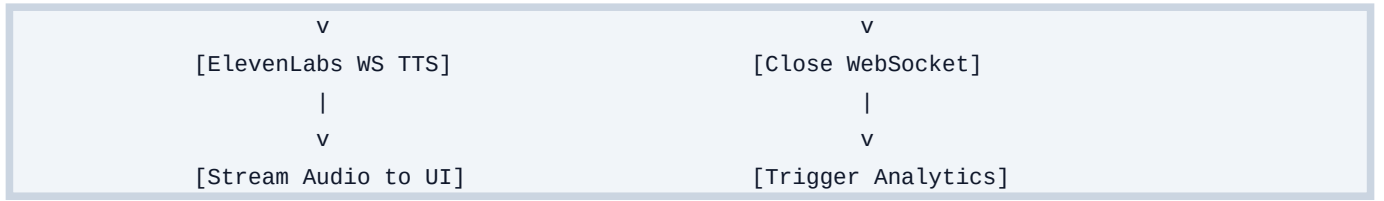
The core of the simulation runs over a stateful WebSocket connection. This ensures low-latency, bidirectional communication for voice activity. The backend manages the conversation loop, responder selection, transcription, and speech synthesis.

The WebSocket Turn Sequence

- **1. Audio Chunk Streaming:** The client streams raw microphone binary audio chunks (base64) over WebSocket to '/ws/live-conversation/{meeting_id}'.
- **2. Whisper Transcription:** The backend accumulates these chunks and calls the OpenAI Whisper service to transcribe the audio text in real-time.
- **3. Live End Phrase Check:** The transcribed text is scanned for goodbye words (bye, see you, thank you, end). If an end is requested, the system skips the LLM generation, speaks a preset goodbye immediately, and completes the meeting.
- **4. Intelligent Responder Selection:** If continuing, GPT-4o-mini is called to identify which company representative (e.g. CTO vs. CFO) should naturally answer based on the context.
- **5. LLM Token Streaming:** The selected representative's persona prompt is loaded, and a token-by-token response is streamed from OpenAI.
- **6. ElevenLabs WebSocket TTS:** These tokens are piped directly into ElevenLabs via a streaming WebSocket connection. This synthesizes voice on-the-fly, reducing delay by 70% compared to traditional HTTP TTS.
- **7. Audio Playback:** The frontend plays the streaming audio bytes sequentially and disables the microphone while the AI is speaking.

Live Voice Flow Pipeline





6. Sample Live Simulation Transcript

To understand how the personas and sales methodology (MEDDIC in this case) interact during a live training session, here is a transcription of a realistic 1-on-2 conversation simulation.

Scenario Parameters

- **Salesperson:** Selling 'OptiTrack Sales Intelligence SaaS'
- **Company:** BioPharm Solutions Inc. (Revenue: \$45M, Size: 250 employees)
- **Representatives:** Bob (VP of Sales - analytical, decision maker) and Alice (CFO - cold_hearted)

Conversation Log

Salesperson (You):

"Hi Bob, hi Alice. Thanks for getting on the call. I wanted to show you how OptiTrack uses predictive analytics to boost pipeline conversion rates. We typically see a 15% increase in deals closed."

Bob (VP of Sales - Analytical):

"Look... everyone says they have predictive analytics. How does your algorithm handle complex, multi-stage enterprise medical sales? Our sales cycles are 9 months long, and we sell to hospitals, not tech companies. What specific metrics can you share about similar healthcare clients?"

Salesperson (You):

"We actually work with MedVance Corp. We integrated our platform with their custom CRM, and we helped them uncover pipeline bottlenecks, reducing their sales cycle from 9 months to 7.2 months."

Alice (CFO - Cold-hearted):

"Reducing the cycle is fine, but it sounds like a costly integration. What is the total cost of ownership including custom integration fees, and how long before we break even on the licensing costs? I need numbers, not generic value statements."

Salesperson (You):

"Typically, licensing is \$100 per user per month. The integration fee is a flat \$15,000, and most clients see full payback within 90 days. We can configure a pilot project first to prove this out."

Bob (VP of Sales):

"A pilot sounds reasonable, but we'd need to agree on clear success criteria before signing anything. Anyway, I have to hop to another meeting. Let's touch base next week. Thanks for the breakdown."

Salesperson (You):

"That sounds great. Thank you, Bob. Goodbye!"

System (Immediate Exit Triggered):

"[Meeting Status: Completed. Total duration: 3.5 minutes. Full audio session recording compiled and uploaded to S3.]"